

Answers

VARC

1.A	2.A	3.D	4.A	5.D	6.D	7.B	8.B
9.C	10.D	11.D	12.D	13.C	14.C	15.C	16.D
17.3142	18.D	19.2431	20.2143	21.D	22.B	23.D	24.C

LRDI

25.84	26.40	27.A	28.D	29.A	30.35	31.38	32.D
33.A	34.A	35.D	36.D	37.A	38.A	39.C	40.40
41.D	42.A	43.C	44.50				

Quant

45.20	46.14	47.4	48.A	49.D	50.B	51.A	52.C
53.12	54.24	55.10	56.D	57.A	58.A	59.B	60.6
61.D	62.A	63.47	64.D	65.D	66.C		

Explanations

VARC

1. A

The author discusses camouflaging techniques in an octopus vis-a-vis other organisms such as cuttlefish, squids and polar bears: { For instance, the use of these cells is how an octopus can wrap itself over a rock and appear jagged or how a **squid or cuttlefish** can imitate the look of a coral reef by growing miniature towers on its skin. It actually matches the texture of the substrate it chooses...Finally, the leucophores: According to a paper, published in Nature, cuttlefish and octopuses possess an additional type of reflector cell called a leucophore. They are cells that scatter full spectrum light so that they appear white in a similar way that a **polar bear's fur** appears white}

However, note that no such discussion on sea snails is presented; hence, Option A is the correct choice.

[VIDEO SOLUTION](#)

2. A

{But they don't look like their cousins at all. Other molluscs include sea snails, sea slugs, bivalves - most are shelled invertebrates with a dorsal foot. Cephalopods are all arms, and can be as tiny as 1 centimetre and as large at 30 feet. Some of them have brains the size of a walnut, which is large for an invertebrate. . . .It makes sense for these molluscs to have added protection in the form of a higher cognition; they don't have a shell covering them, and pretty much everything feeds on cephalopods, including humans.}

We can understand that B, C and D are true - they are points of dissimilarities between octopuses and other molluscs [we are told that they have multiple appendages instead of a single dorsal foot and lack shells; they also have higher intelligence]. However, there is no information on whether humans consume molluscs like sea snails or not [furthermore, the author does not use this fact to claim that octopuses are distinct in this regard].

Hence, Option A is the correct choice.

 VIDEO SOLUTION

3. D

Options A and C: The discussion on Chromatophores, iridophores and leucophores sufficiently supports the statements here.

Option B: We can infer this from the following excerpt - *{Papillae are sections of the skin that can be deformed to make a texture bumpy... For instance, the use of these cells is how an octopus can wrap itself over a rock and appear jagged or how a squid or cuttlefish can imitate the look of a coral reef by growing miniature towers on its skin. It actually matches the texture of the substrate it chooses.}*

Option D is not discussed anywhere in the passage.

 VIDEO SOLUTION

4. A

Let us evaluate the choices individually:

Option A: [the number of chromatophores in Cephalopods is half the number of iridophores and leucophores] If true, this does not undermine the camouflaging adeptness of Cephalopods primarily because each of chromatophores, iridophores and leucophores have specific [somewhat independent] roles to play and it is unclear how their quantity would directly impact the camouflaging capacity. Even if the number of chromatophores is fewer than the other types, the octopus can still maintain its camouflaging adeptness.

Option B: [the temperature of water at the depths at which Cephalopods reside renders the transmission of neural signals difficult.] If true, this would limit the camouflaging adeptness primarily because the underlying mechanism is being restricted/impacted. {When the cephalopod wants to change colour, the brain carries an electrical impulse through the nerve to the muscles that expand outwards, pulling open the sacs to display the colours on the skin.}

Option C: [light reflects the colours red, green, and yellow at the depths at which Cephalopods reside.] If true, this would limit the camouflaging adeptness primarily because the underlying mechanism is being restricted/impacted. If the colour scheme is distinct, it would undermine the observations presented below: *{[Chromatophores] are organs on their bodies that contain pigment sacs, which have red, yellow and brown pigment granules...Why these three colours? Because these are the colours the light reflects at the depths they live in (the rest is absorbed before it reaches those depths)}*

Option D: [the hydrostatic pressure at the depths at which Cephalopods reside renders radial muscle movements difficult.] If true, this would limit the camouflaging adeptness primarily because the underlying mechanism is being restricted/impacted. {These sacs have a network of radial muscles, meaning muscles arranged in a circle radiating outwards.}

Hence, Option A is the correct choice.

 VIDEO SOLUTION

5. D

Based on the passage, it is likely that the author would approve of all of the options except for Option D, which is moving towards technical-social dualism. Technical-social dualism is described in the passage as the idea that the technical and social dimensions of engineering problems are readily separable and remain distinct throughout the problem-definition and solution process. The passage criticizes this approach, arguing that it ignores the social dimensions of engineering problems and leads to a focus on technical ideals such as cost and efficiency at the expense of broader societal concerns. Therefore, it is unlikely that the author would approve of moving towards technical-social dualism.

Option A is likely to be included because the passage mentions the need for engineers to be aware of the potential impacts of their work on different groups of people, including the environment. It is suggested that ignoring these factors can result in technologies that are not sustainable, which can contribute to a crisis of medical waste and health care, accounting for 10% of U.S. greenhouse gas emissions.

Option B is likely to be included because the passage discusses the consequences of ignoring social dimensions in engineering, such as physical failures and the perpetuation of social inequities. It suggests that a more responsible approach to technical design and problem-solving would consider the full range of stakeholders and the potential impacts of a technology on different groups of people.

Option C is likely to be included because the passage emphasizes the importance of considering social justice in engineering education and practice. It mentions courses focusing on place-based knowledge and community engagement as examples of efforts to incorporate social justice into engineering education. Such an approach would involve designing technologies that are responsive to the needs of communities, using local knowledge and taking into account local priorities.

Hence, Option D is the correct choice.

 VIDEO SOLUTION



6. D

Option D is not mentioned in the passage as a negative outcome of focusing on technical ideals in the medical sphere. The passage specifically mentions that "most FDA-approved drugs are incorrectly dosed for people assigned female at birth, leading to unexpected adverse reactions. This is because they have been inadequately represented in clinical trials," which suggests that the incorrect dosing of drugs for people assigned female at birth is a consequence of inadequate representation in clinical trials, rather than a result of focusing on technical ideals.

The other options are all mentioned in the passage as negative outcomes of focusing on technical ideals in the medical sphere. Option A refers to the passage's mention of the lack of technologies for "systemically marginalized people" such as those with endometriosis. Option B relates to the discussion on spirometers that have correction factors that assume smaller lung capacity in Black and Asian individuals based on research by eugenicists. Option C is tied to the discussion on "most FDA-approved drugs" being incorrectly dosed for people assigned female at birth due to inadequate representation in clinical trials.

Hence, Option D is the correct choice.

7. B

The passage discusses the concept of technical-social dualism, which separates the technical and social dimensions of engineering problems and can result in a focus on technical ideals such as cost and efficiency at the expense of broader societal concerns. The passage states that this way of teaching "primes students to believe that their decision-making is purely objective, as it is grounded in math and science" and that "nontechnical parameters such as access to a technology, cultural relevancy or potential harms are deemed political and invalid." These statements support the claim that engineering students are taught to focus on objective technical outcomes, independent of the social dimensions of their work. Option B best captures the above understanding. The other answer choices do not accurately reflect the content or the main argument of the passage.

8. B

Option A: [Correct] The passage states that technical ideals, such as cost and efficiency, are often determined by a dominant culture that prioritizes economic growth for the most privileged segments of society. This can result in technologies and interventions that are not designed with the needs and concerns of marginalized groups in mind, leading to systemic discrimination against these groups.

Option B: [Incorrect] The passage does not mention anything about sustainability or medical waste contributing to greenhouse gas emissions as a reason for the systematic discrimination of marginalized people in technology-related interventions.

Option C: [Correct] The passage mentions that certain technologies, such as spirometers, have correction factors that assume smaller lung capacity in Black and Asian individuals based on research by eugenicists who believed in racial hierarchies and considered nonwhite people as inferior. This is an example of how subjective beliefs can be treated as facts and encoded into technologies, leading to social inequities.

Option D: [Correct] The passage discusses how subjective beliefs treated as facts by those in decision-making roles can result in physical failures, such as incorrect dosing of drugs for people assigned female at birth, and also encode social inequities, such as the correction factors on spirometers that assume smaller lung capacity in Black and Asian individuals.

Hence, Option B is the correct choice.

9. C

The author states that the stock exchange and the black market are both market institutions, but notes that the stock exchange is formal, while the black market is not. This suggests that the two institutions differ in terms of how they are structured and operate, and the author suggests that this difference is an example of how different kinds of institutions can co-exist within the same domain. Option C aptly presents this idea. Contrarily, the ideas mentioned in Options A, B and D are not supported by the information in the passage.

Hence, Option C is the correct choice.

10. D

Based on the passage, the correct inference is: Option D - **"natural language" refers to that stage of language development where no conscious human intent is evident in the formation of language.**

The passage states that natural languages are "the result of human action, but not the execution of any human design," suggesting that they are not consciously designed or modified. Artificial languages and reformed natural languages, on the other hand, are described as introducing "design into their modifications or refinements," indicating that they are consciously modified.

The passage does not support the other inferences. In fact, it suggests that culture and tradition can be understood as subsets of institutions and have analytical, explanatory, and expository power when they are studied in this context. It also suggests that there are both informal and formal institutions, and that many institutions are a mixture of the two, rather than being mutually exclusive categories. The passage also does not imply that institutions like the family, rituals, governance, economy, and the military are natural and cannot be consciously modified; rather, it suggests that these institutions are the product of conscious attempts to mold and plan them, and that they can be modified through processes such as family law, established and disestablished churches, constitutions and laws, and so on [Options A, B and C].

CAT Percentile Predictor



11. D

{We begin with the emergence of the philosophy of the social sciences as an arena of thought and as a set of social institutions. The two characterisations overlap but are not congruent. Academic disciplines are social institutions. . .}

In the first paragraph of the passage, the two "characterisations" that are seen as overlapping but not congruent are the philosophy of the social sciences as an arena of thought [academic discipline] and as a set of social institutions. The author suggests that these two characterizations overlap in the sense that they both involve the study of social phenomena, but they are not congruent because the philosophy of the social sciences is a field of study that encompasses a broad range of ideas and theories, while social institutions are specific structures that organize and coordinate social action. Although they are related, they are not identical and can be understood as distinct but overlapping aspects of the social sciences. Option D correctly represents the above.

12. D

The passage discusses the concept of institutions within the philosophy of the social sciences. Institutions are seen as structures that coordinate the actions of individuals and can be either formal or informal; in this regard, the author discusses various types of institutions, including universal institutions, formal institutions, and informal institutions. The author also argues that culture and tradition can be seen as subsets of institutions, and that language is a particularly important institution that shapes social life and science. The author notes that there can be differences in the way that institutions are understood and

emphasized by different social scientists but that these differences are often matters of emphasis rather than fundamental disagreements. Primarily, the passage underlines that these institutions are structures that coordinate the actions of individuals. Option D best captures the essence of the abovementioned elements.

Option A is incorrect because it only partially reflects what the author says about culture and tradition. While the author does discuss culture and tradition in relation to institutions, they are not treated as being completely separate from institutions. Instead, the author argues that culture and tradition can be seen as subsets of institutions, meaning they are part of institutions rather than completely separate from them. Option B exaggerates the emphasis the author places on language as an institution. While the author does discuss language as an important institution, he does not suggest that it is the only or even the most fundamental institution. Option C, while true, does not capture the main focus of the passage, which is on the concept of institutions in general rather than on specific examples of institutions.

[VIDEO SOLUTION](#)

13. C

The passage is about the idea that making music is a fundamental and universal aspect of the human experience, and that it has a long history that is both sociocultural and biological in nature. The author suggests that the capacity for musicking, or making music, is innate in humans and is closely related to other capacities such as language and symbol-making. The author also acknowledges that there are variations in how musical capacities are expressed and developed among different cultures, but maintains that all humans possess these capacities to some extent. The author suggests that the emergence of music can be traced back to at least 50,000 years ago, and that it likely developed gradually over a longer period of time. The author also notes that the emergence of music involved the recruitment of many cognitive capacities, and that understanding this process involves following multiple strands. In this regard, Option C includes all the keywords central to the discussion.

The other choices do not include all of the relevant terms. Option A includes some of the terms mentioned in the passage, but omits important ones, such as humans and modern humanity. Option B includes humans and capacities but omits important terms such as musicking and symbol-making. Option D includes humans and the emergence of music but omits important terms such as musicking and linguistic capacities.

Hence, Option C is the correct choice.

[VIDEO SOLUTION](#)

14. C

In this context, the author is trying to communicate that although there may be various qualifications and considerations that might trail after the statement that "humans today make music," the statement is fundamentally true. The author suggests that almost all humans are musicians to some extent, and that the capacity for making music is a fundamental aspect of the human experience. The author is urging readers to consider this statement without getting bogged down in the various qualifications and considerations that might be attached to it, and to recognize its underlying truth. Option C aptly captures this idea.

Option A: [Incorrect] The phrase "trail after" does not necessarily imply that a bald statement is followed by a series of qualifying clarifications and caveats. Rather, it simply means that something follows after something else. In this case, the author is suggesting that there may be various qualifications and considerations that follow after the statement that "humans today make music," but is not implying that the statement itself is trailed by a series of clarifications and caveats.

Option B: [Incorrect] The phrase "bald statement" does not necessarily imply that a statement requires no qualifications to infer its meaning. Rather, it simply means that the statement is presented in a straightforward and unembellished way. In this case, the author is suggesting that the statement that "humans today make music" is presented in a bald and straightforward way, but is not implying that the statement itself requires no qualifications or considerations.

Option D: [Incorrect] The phrase "give free reign to" does not accurately describe the author's intention in this context. The author is not suggesting that readers should allow musical expressions to be unrestricted or uncontrolled, but rather that they should consider the statement that "humans today make music" without getting bogged down in various qualifications and considerations.

Hence, Option C is the correct choice.

[VIDEO SOLUTION](#)

15. C

Option A: This is incorrect because it overgeneralizes the ideas presented in the passage. The passage does not state that all humans are musicians or capable of musicking, only that making music is a universal aspect of the human experience and that the capacity for musicking is innate in humans.

Option B: The passage does state that musicking is a form of expression, but there is no discussion supporting the claim "musicking is not language-like." Thus, we can eliminate this choice.

Option C: The passage states - ("...if we look back 20,000 years, a small portion of this long period, we reach the lives of humans whose musical capacities were probably little different from our own...") This suggests that the musical capacities of humans 20,000 years ago were not significantly different from those of humans today.

Option D: This is not supported by the passage. The passage does not state that all musical work is located in the overlap between linguistic capacity and music production. In fact, the passage states that "most of these capacities overlap with nonmusical ones, though a few may be distinct and dedicated to musical perception and production," suggesting that there are some capacities specifically dedicated to musical perception and production that do not overlap with nonmusical ones.

Hence, the correct answer is Option C.

[VIDEO SOLUTION](#)



16. D

Let us examine the given statements:

Option A: [Nonmusical capacities are of far greater consequence to human survival than the capacity for music.] It is unclear how this ties into the discussion since the survival aspect is not touched upon or implied in the passage. Thus, the claim, if true, does little to undermine the author's claim that humans are musicking creatures.

Option B: [From a cognitive and psychological vantage, musicking arises from unconscious dispositions, not conscious ones.] This does not weaken the author's claim that musicking is a universal aspect of the human experience - while the author does suggest that musicking arises from innate dispositions, the fact that these dispositions may be unconscious rather than conscious does not weaken the overall argument.

Option C: [As musicking is neither language-like nor symbol-like, it is a much older form of expression.] This, again, does not contradict the author's claim in the passage. Though the author suggests that musicking is distinct from language and symbol-making, this does not necessarily mean that it is a much older form of expression. In fact, the author notes that the emergence of musicking can be traced back to at least 50,000 years ago, which is relatively recent in evolutionary terms. Even if this were true, it does not undermine the author's claim.

Option D: [Musical capacities are primarily socio-cultural, which explains the wide diversity of musical forms.] If true, this directly contradicts the author's claim that musicking is a universal aspect of the human experience. The author argues that the capacity for musicking is innate in all humans and that it has a long history that is both sociocultural and biological in nature. However, if musical capacities are primarily socio-cultural, as suggested in Option D, this would mean that musicking is largely shaped by cultural and social factors rather than being a fundamental aspect of the human experience. This would greatly weaken the author's overall argument that all humans are musicking creatures. Hence, Option D is the correct choice.

VIDEO SOLUTION

17. 3142

The given set of sentences talks about the concept of cheerfulness and how it can affect a person's internal state. It discusses the history of the word "cheer," which originally referred to a person's facial expression, and how it came to be associated with an abstract concept of positive emotion. The passage also notes that translations of the Bible into vernacular languages helped to expand the meaning of "cheer" to include a sense of cheerfulness or positivity as a quality of the self and a moral community. We note that sentences 1, 2 and 4 are part of the timeline the author traces, while statement 3 is a broad claim that would serve as an apt introduction. The idea of cheerfulness and self that is mentioned in 3 is linked to the "trajectory of cheerfulness through the self" highlighted in statement 1, allowing us to form the pair [3-1]. Note that 4 discusses how cheer came to be perceived as a trait at the intersection of emotion and a physical feature [as highlighted in 1 - where it has been categorised as a facial feature]. Sentence 2 then extends on this idea by how cheerfulness, over time, came to be considered as "an emotional and social quality." Hence, the correct arrangement is 3142.

VIDEO SOLUTION

18. D

The sentence "**Most were first-time users of a tablet and a digital app**" would fit best in blank 4 because it provides information about the specific group of people who were involved in the pilot study for the app. The preceding sentence mentions that the app was piloted in early 2021 with over 400 women from rural areas, and the added sentence provides further detail about the specific experiences that these users had with technology. This information helps to clarify that the pilot study for the app involved a group of people who may be unfamiliar with digital devices and services, and it also helps to emphasize the potential impact that the app may have had on these users.

Placing the sentence in blank 4 is logical because it provides relevant and specific information about the pilot study for the app, which is the topic being discussed in that part of the paragraph. By contrast, placing the sentence in any of the other blanks would be less logical because it would be introducing information that is not directly related to the main topic being discussed in those parts of the paragraph.

Hence, Option D is the correct choice.

VIDEO SOLUTION

19. 2431

The statements here discuss how the way we think about food often centres around individual choice, rather than acknowledging the larger societal and structural factors that influence food choices. It also mentions the role of neoliberalism in shifting the responsibility for food onto individual consumers and how this may disproportionately impact women, who may feel a sense of responsibility to navigate and "mediate" an unhealthy food system. Sentences 2 and 4 create a logical block since 2 highlights the framing of food through the "lens of individual choice", and 4 extends on the impact of such a framing ["appeals to a popular desire to experience agency, but draws away from the structural obstacles that stratify individual food choices"]. "This" in statement 3 refers to the particular outcome of framing underlined in 4, allowing us to create a logical block: [2-4-3]. Statement 1 can be placed at the end of this arrangement to emphasise how women are perhaps disproportionately impacted by the elements discussed in 2-4-3. Hence, the correct answer is 2431.

VIDEO SOLUTION

20. 2143

The statements discuss the complexity of environmental issues and the need for political solutions rather than technological ones to address the threat of environmental collapse. The discussion focuses on the various causes of biodiversity loss and ecosystem collapse, such as chemical pollutants, damming of rivers, and invasive species, and highlights that there is no single technological solution to address these issues. The suggestion is that we should consider moving towards stable-state economies, which preserve markets and competition while also recognizing the rights of future generations and other species and allowing for stronger environmental protections. Statements 1 and 2 both address the complexity and diverse nature of environmental issues. Statement 2 notes that while discourse often focuses on emissions, there are many other factors contributing to environmental collapse, including biodiversity loss and ecosystem collapse. Statement 1 then adds to this by outlining specific examples of diverse causes of environmental problems, including chemical pollutants, damming of rivers, and invasive species. Statements 4 and 3 offer suggestions to combat the issues highlighted in 2 and 1. Hence, the correct arrangement is 2143.

VIDEO SOLUTION



21. D

The passage is about the 'inherent vice' or the natural tendency of certain artworks to deteriorate over time due to various factors such as discolouration of pigments, cracking of varnishes, and warping of canvases. The passage also mentions an example of Damien Hirst's artwork, The Physical Impossibility of Death in the Mind of Someone Living, which began to decompose quickly due to the use of the wrong preserving liquid. In this regard, Option D offers an apt summary of the passage because it accurately captures the main idea of the passage, which is that artworks may not last forever and may deteriorate with time. Option A is incorrect because the passage does not mention any moral responsibility of museums to restore and preserve artworks. Similarly, Option B can be eliminated since the passage does not specifically mention museums guarding art treasures from intrinsic defects. Option C is also inaccurate because the discussion does not present the evolution of the role of museums in preserving artworks forever.

Hence, Option D is the correct choice.

VIDEO SOLUTION

22. B

The passage discusses the ways in which big data and targeted online content can potentially influence and manipulate behaviour, leading to concerns over freedom and privacy in the digital age. This is reflected in the statements that "behavioural control" in the age of big data echoes Cold War-era anxieties about "brainwashing" and "repression," and that the use of targeted online content can enable "influence to take place on a pre-reflexive level." The passage also mentions the fear that the digital age has not liberated us, but rather exposed us by making personal and behavioural data available to machine-learning algorithms. Option B accurately reflects this central theme of the passage by stating that the debate on the nature of freedom and privacy has resurfaced due to the availability of personal information through big data. Option A is incorrect because it goes beyond the scope of the passage by stating that digital technology is "enslaving" us, which is not explicitly stated in the text. Similarly, Option C is inaccurate since the author only mentions the Cold War as a reference point for similar debates on behavioural control, but does not focus on the Cold War itself. Option D is wrong because the passage does not mention artificial intelligence specifically, but rather machine-learning algorithms.

Hence, Option B is the correct choice.

VIDEO SOLUTION

23. D

The passage is about the political and civic organization of ancient cities. It states that some ancient cities were organized along egalitarian lines, without any indication of temples or palaces (which suggests a lack of a ruling class or administrators), and that in other cities, temples and palaces never emerged at all. Option D correctly summarizes this information by stating that there was evidence of an egalitarian urban life in some ancient cities, where the political and civic organization was less hierarchical. Option A is incorrect because it only mentions the absence of temples and palaces, but does not mention the fact that some ancient cities were organized along egalitarian lines. On a similar note, Option B presents an exaggeration by suggesting that 'all' ancient cities were organized along egalitarian lines, which is not stated in the passage. Option C is also inaccurate since it asserts that ancient cities were transformed from egalitarian to hierarchical, but the passage only states that some ancient cities were egalitarian and does not mention any transformation.

Hence, Option D is the correct choice.

VIDEO SOLUTION

24. C

We can easily eliminate Options A and B since the given sentence "This was years in the making but fast-tracked during the pandemic, when people started being more mindful about their food", he explained" uses the pronoun "he," which must refer to "Karmakar." The sentence should take the spot of blank 3 since it fits in well with the idea that ghee is part of a "return-to-basics movement in India", as it mentions that the renewed interest in ghee has been developing for a while but was accelerated during the pandemic when people became more conscious about their food choices. Additionally, the idea that this movement is part of an overall trend towards "slow food" is also mentioned in the sentence, as it talks about people being mindful of their food. By contrast, placing the sentence in any of the other blanks would be less logical because it would be introducing information that is not directly related to the main topic being discussed in those parts of the paragraph.

Hence, Option C is the correct choice.

VIDEO SOLUTION

LRDI

Explanation [25 - 29]:

In statement 1, it is given that all three of them met the same total number of households, and each of them sold a total of 100 items in two days. In statement 2, it is given that on both days, Lahur met the same number of households and sold the same number of items. This implies he sold 50 items per day. Let the number of households Lahur met in a day be 'x'.

Total number of households each of them met in two days will be '2x'.

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri					2x	100
Hokli					2x	100
Lahur	x	50	x	50	2x	100

In statement 3, it is given that Hokli could not sell any item on the second day because the first household he met on that day complained against him. This implies he met only 1 household on day 2.

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri					2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 4, it is given that Tohri met 30 more households on the second day than on the first day.

Let the number of households Tohri met on day 1 be 'a'

It is given,

$$a + a + 30 = 2x$$

$$a + 15 = x$$

$$a = x - 15$$

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri	x-15	y	x+15	100-y	2x	100
Hokli	2x-1	100	1	0	2x	100
Lahur	x	50	x	50	2x	100

In statement 5, it is given that

$$2 \left(\frac{50}{x} \right) = \frac{y}{x-15} \dots\dots (1)$$

$$\frac{3}{4} \left(\frac{50}{x} \right) = \frac{100-y}{x+15} \dots\dots (2)$$

$$\frac{100}{x} = \frac{y}{x-15}$$

$$\frac{100}{y} = \frac{x}{x-15}$$

$$\frac{y}{100} = 1 - \frac{15}{x}$$

$$x = \frac{1500}{100-y}$$

Substituting x in (2), we get

$$y = 40 \text{ and } x = 25$$

Final Table:

	Day 1		Day 2		Total	
	HHs	Items	HHs	Items	HHs	Items
Tohri	10	40	40	60	50	100
Hokli	49	100	1	0	50	100
Lahur	25	50	25	50	50	100

25. **84**

The total number of households met by Tohri, Hokli and Lahur on the first day is $10 + 49 + 25$, i.e. 84.

[VIDEO SOLUTION](#)



26. **40**

The number of items sold by Tohri on the first day is 40.

[VIDEO SOLUTION](#)

27. **A**

Lahur met 25 households on day 2. The answer is option A.

[VIDEO SOLUTION](#)

28. **D**

Tohri met 10 households on day 1. The answer is option D.

[VIDEO SOLUTION](#)

29. **A**

Among the three, Tohri had the highest success rate on the second day - this statement is incorrect. On day 2, Lahur had the highest success rate, i.e. 2 whereas Tohri's success rate is 1.5.

Tohri had a higher success rate on the first day compared to the second day - this statement is correct.

Tohri's day 1 success rate is 4 and day 2 success rate is 1.5.

Among the three, Tohri had the highest success rate on the first day - this statement is correct.

Tohri's success rate on day 1 is 4.

Hokli's success rate on day 1 is 2.04.

Lahur's success rate on day 1 is 2.

Among the three, Lahur had the lowest success rate on the first day - this statement is correct.

Tohri's success rate on day 1 is 4.

Hokli's success rate on day 1 is 2.04.

Lahur's success rate on day 1 is 2.

The answer is option A.

[VIDEO SOLUTION](#)

30. **35**

Points to be noted:

1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.

2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.

3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse (Prefers minimum distance).

In the question, it is given that last location is A. The demand in the remaining places should be greater than A. This implies A demand cannot be 70 units. Therefore, it is 50 units.

The demand of the location D is 30 or 50 units. This implies this should be placed before D.

The demand of the location placed before D should be greater than or equal to 50 units. Location supplier visited before D is B (60 units of demand). It cannot be C because values of C is greater than the values of B.

Therefore, order is C - B - D - A.

From warehouse to C - 12 km

C to B - 4 km

B to D - 12 km

D to A - 7 km (through warehouse)

Total distance covered = $12 + 4 + 12 + 7 = 35$ km

This question is removed from the paper because if the order is CBDA, the supplier will go to A from B (and hence the last city visited will be D).



31.38

Points to be noted:

1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse (supplier prefers minimum distance).

In the question, it is given that total number of units delivered is 250 units.

Maximum number of widgets that can be delivered is 70 units(A) + 50 units(D) + 60 units(B) + 100 units(C) = 280 units

From this 30 units should be decreased. 30 units can be decreased only when C's demand decreases to 70 units (because the difference for remaining locations is 20 units)

Therefore, the only possibility is 70 units(A) + 50 units(D) + 60 units(B) + 70 units(C)

From statement 1, the order should be A - C - B - D (as A is nearer to warehouse than C)

Distance from Warehouse to A is 5 km

Distance from A to C is 17 km

Distance from C to B is 4 km

Distance from B to D is 12 km

Total distance covered = 5 + 17 + 4 + 12 = 38 km

32.D

Maximum number of widgets delivered in a day is 100 units(C) + 70 units(A) + 60 units(B) + 50 units(D), i.e. 280 units

Given, total number of widgets delivered is 260 units. This implies 20 units must be decreased from any one of the locations.

A - (70,50), B - (60,40), C - (100,70) and D - (50,30)

20 units can be decreased from A, B or D.

Demand at location C will be 100 units and supplier first visits C.

In the question, it is also given that the route ends at B.

C(100 units), $\rightarrow$ $\rightarrow$ B

If B's demand is 60 units, D's demand should be more than 60 units which is not possible. Therefore, B's demand should be 40 units.

20 units is decreased at location B. This implies demand at location A is 70 units and at location D is 50 units.

Order will be C(100 units) - A(70 units) - D(50 units) - B(40 units).

It is given,

C - 100 units - 70%

A - 70 units - 60%

D - 50 units - 60%

B - 40 units - 30%

Required value = $0.7 \times 0.6 \times 0.6 \times 0.3 = 0.0756 = 7.56\%$

The answer is option D.

33.A

It is given that the first location visited from the warehouse is A.

If A's demand is 50 units, C's demand should be less than 50 units which is not possible. This implies demand of locations A and C is 70 units.

A(70 units) $\rightarrow$ C(70 units)

Warehouse to A - 5km

A to C - 17 km

Distance covered = 5 + 17 = 22 km

Remaining distance = 40 - 22 = 18 km

C to B - 4 km

B to D - 12 km

Distance covered = 4 + 12 = 16 km $\neq$ 18 km

C to D - 6 km

D to B - 12 km

Distance covered = 6 + 12 = 18 km

Therefore, supplier can cover distance 18 km if he visits D before B, i.e. demand of D should be more than demand of B. This is only possible when D's demand is 50 units and B's demand is 40 units.

It is given,

D - 50 units - 60% probability

B - 40 units - 30% probability

Required value = $0.6 \times 0.3 = 0.18 = 18\%$

The answer is option A.

[VIDEO SOLUTION](#)

34. A

Points to be noted:

1. Starts from the warehouse and ends at a location after visiting all the locations exactly once.
2. While making the route plan, the supplier goes to the locations in decreasing order of demand. If equal demand, goes to the nearest ones first.
3. While creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse (Prefers minimum distance).

Demand in all other locations should be less than or equal to the demand in first location. In the question, it is given that A is not first location. B and D cannot be first location. This implies C should be the first location.

It is given, total route distance is 29 km.

Warehouse to C is 12 km.

This implies remaining distance should be $29 - 12$, i.e. 17 km

This is only possible when C visits B, A and D later ($4 + 6 + 7$).

The order should be C - B - A - D

C's demand can be 70/100, B's demand should be 60, A's demand should be 50 and D's demand can be 30/50.

The possible number of widgets delivered can be

$70 + 60 + 50 + 30 = 210$

$70 + 60 + 50 + 50 = 230$

$100 + 60 + 50 + 30 = 240$

$100 + 60 + 50 + 50 = 260$

The answer is option A.

[VIDEO SOLUTION](#)

35. D

It is given,

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

Company A:

Revenue = 240 and cost incurred = 180

Profit = $240 - 180 = 60$

Company B:

Revenue = 220 and cost incurred = 145

Profit = $220 - 145 = 75$

Company C:

Revenue = 195 and cost incurred = 110

Profit = $195 - 110 = 85$

Company D:

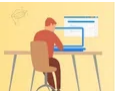
Revenue = 140 and cost incurred = 160

No profit.

Company C had the highest annual profit.

The answer is option D.

[VIDEO SOLUTION](#)



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CAT Sectional Tests



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36. D

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

For all the companies in all three years, cost incurred is less than Revenue except for D in 2020.

Revenue is 20 and cost incurred is 50

Company D experienced the highest annual loss in 2020.

The answer is option D.

[VIDEO SOLUTION](#)

37. A

It is given,

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

Measure of A's performance in 2019 = $\frac{90-85}{85} = \frac{5}{85} = 0.06$

Measure of B's performance in 2019 = $\frac{100-75}{75} = \frac{25}{75} = 0.33$

Measure of C's performance in 2019 = $\frac{25-20}{25} = \frac{5}{25} = 0.2$

Measure of D's performance in 2019 = $\frac{50-40}{40} = \frac{10}{40} = 0.25$

Company A had the lowest value.

The answer is option A.

[VIDEO SOLUTION](#)

38. A

It is given,

Company	Strength			New hires		
	2019	2020	2021	2019	2020	2021
A	150	140	150	20	35	25
B	210	240	250	35	45	30
C	320	320	320	45	40	35
D	400	410	400	30	35	40

Company A:

The number of employees in the beginning of 2019 = 150

The number of employees hired in 2019 = 20

The number of employees should be at the beginning of 2020 is 150+20, i.e. 170 but there are 140 only. This implies 30 left company A in 2019.

The number of employees in the beginning of 2020 = 140

The number of employees hired in 2020 = 35

The number of employees should be at the beginning of 2021 is 140+35, i.e. 175 but there are 150 only. This implies 25 left company A in 2020.

The number of employees left company A in 2019 and 2020 = 30 + 25 = 55

Company B:

Similarly, the number of employees left company B in 2019 = 210 + 35 - 240 = 5

The number of employees left company B in 2020 = 240 + 45 - 250 = 35

The number of employees left company B in 2019 and 2020 = 5 + 35 = 40

Company C:

Similarly, the number of employees left company C in 2019 = 320 + 45 - 320 = 45

The number of employees left company C in 2020 = 320 + 40 - 320 = 40

The number of employees left company C in 2019 and 2020 = 45 + 40 = 85

Company D:

Similarly, the number of employees left company D in 2019 = 400 + 30 - 410 = 20

The number of employees left company D in 2020 = 410 + 35 - 400 = 45

The number of employees left company D in 2019 and 2020 = 20 + 45 = 65

The total number of employees lost in 2019 and 2020 is least for company B.

The answer is option A.

[VIDEO SOLUTION](#)

39. C

It is given,

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

Company	Strength			New hires		
	2019	2020	2021	2019	2020	2021
A	150	140	150	20	35	25
B	210	240	250	35	45	30
C	320	320	320	45	40	35
D	400	410	400	30	35	40

$$\text{Company A} = \frac{90-65}{145} = \frac{25}{145} = \frac{50}{290}$$

$$\text{Company B} = \frac{90-40}{245} = \frac{50}{245}$$

$$\text{Company C} = \frac{70-60}{320} = \frac{10}{320}$$

Company B had the highest profit per employee.

The answer is option C.

VIDEO SOLUTION

Explanation [40 - 44]:

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products = $320/2 = 160$

Total(32)	Domestic (160)	Cosmetic	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
		Nutrition	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic	EU -	
			FDA -	
Nutrition		EU -		
		FDA -		

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic (90)	EU -	
			FDA -	
Nutrition (70)		EU -		
		FDA -		

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are $130-80$, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is $120-80$, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is $70-40$, i.e. 30.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
Nutrition (70)		EU - 40		
		FDA - 30		

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is $90-80$, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = $70-50 = 20$

Total(32)	Domestic (160)	Cosmetic (90)	only EU - 10	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU - 20	Total EU -
			only FDA -	Total FDA - 50
		Both -		
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
Nutrition (70)		EU - 40		
		FDA - 30		

40.40

The number of foreign, cosmetic and FDA approved products is 40.

The answer is 40.

[VIDEO SOLUTION](#)



41.D

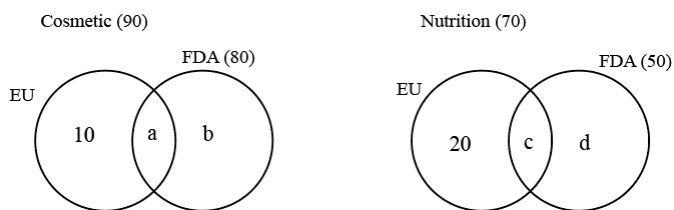
The number of cosmetic products which do not have FDA approval = domestic only EU + foreign EU = 10 + 50 = 60

The answer is option D.

[VIDEO SOLUTION](#)

42.A

In statement 3, it is given that the number of domestic products which have both the approvals = 60



In the question, it is given that $a + c = 60$

To find the minimum value of a, we need to maximise c.

Maximum value c can take is 50

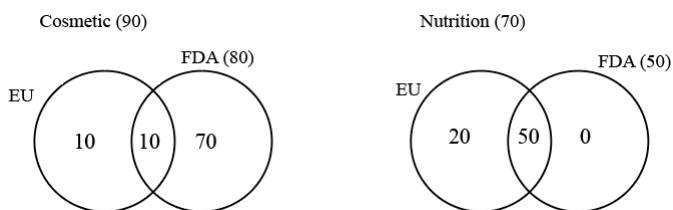
Therefore, minimum value of a is $60 - 50$, i.e. 10.

To find the maximum value of a, we need to minimise c.

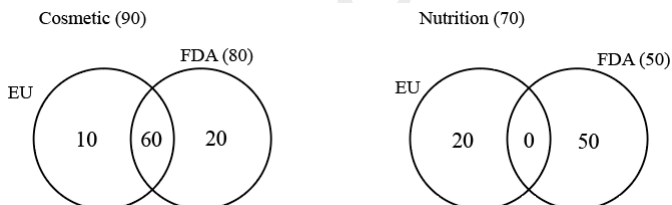
Maximum value c can take is 0.

Therefore, maximum value of a is $60 - 0$, i.e. 60.

a is minimum:



a is maximum:



Therefore, the number of domestic cosmetic products that had both the approvals is at least 10 and at most 60.

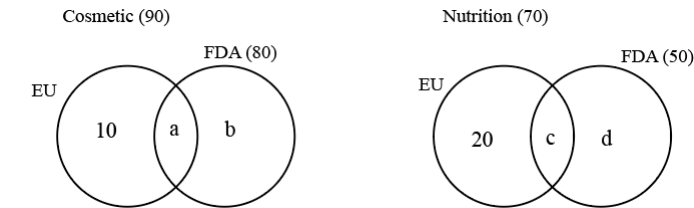
The answer is option A.

[VIDEO SOLUTION](#)

43.C

In the question, it is given that 70 cosmetic products did not have EU approval.

In foreign, 40 cosmetic products did not have EU approval. This implies 30 cosmetic products should have only FDA approval in domestic products.

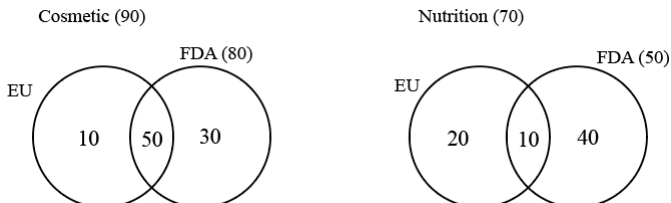


According to the above statement, $b = 30$

$$a = 80 - 30 = 50$$

Given, $a + c = 60$

$$c = 60 - 50 = 10$$



Therefore, the number of nutrition products which had both the approvals is 10.

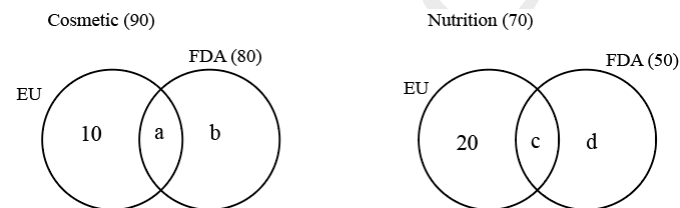
The answer is option C.

[VIDEO SOLUTION](#)

44. 50

In the question, it is given that 50 nutrition products did not have EU approval.

In Foreign products, there are 30 nutrition products which do not have EU approval. This implies 20 nutrition products do not have EU(have only FDA) approval in domestic products.



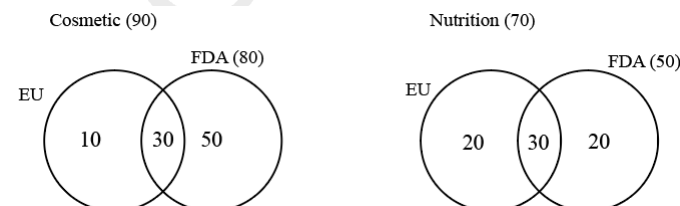
It is given, $d = 20$

$$c = 50 - 20 = 30$$

It is given, $a + c = 60$

$$a = 60 - 30 = 30$$

$$b = 80 - 30 = 50$$



Therefore, the number of domestic cosmetic products did not have EU(only FDA) approval is 50.

[VIDEO SOLUTION](#)

Quant

45. 20

Let the total investment be $15x$ and the no. of years required be T years

$$\frac{(3x \times 6 \times T)}{100} + \frac{(5x \times 10 \times T)}{100} + \frac{(7x \times 1 \times T)}{100} \geq 15x$$

$$\text{or, } \frac{75xT}{100} \geq 15x$$

$$\text{or, } T \geq 20$$

So minimum value of T is 20 years

[VIDEO SOLUTION](#)

46.14

$$a_1 + a_2 + \dots + a_N = 300N$$

$$6a_1 + a_2 + \dots + a_N = 400N$$

$$5a_1 = 100N$$

$$a_1 = 20N$$

As the given sequence of numbers is non-decreasing sequence, N can take values from 2 to 15.

N is not equal to 1, if N = 1, then average of N numbers is 300 wouldn't satisfy.

Therefore, N can take values from 2 to 15, i.e. 14 values.

VIDEO SOLUTION

47.4

Case 1: When $x^2 - 3x - 10 = 0$ and $x^2 - 10 \neq 0$

$$x^2 - 3x - 10 = 0 \text{ or, } (x - 5)(x + 2) = 0$$

or, $x = 5$ or -2

Case 2: $x^2 - 10 = 1$

$$x^2 - 11 = 0$$

No integer solutions

Case 3: $x^2 - 10 = -1$ and $x^2 - 3x - 10$ is even

$$x^2 - 9 = 0$$

$$\text{or, } (x+3)(x-3)=0$$

or, $x = -3$ and 3

for $x = -3$ and $+3$ $x^2 - 3x - 10$ is even

In total 4 values of x satisfy the equations

VIDEO SOLUTION

48.A

Savings target in a year = $550 \times 12 = \text{Rs } 6600$

Saving in first 9 months = $9(4000 - 3500) = \text{Rs } 4500$

Saving for remaining 3 months should be $6600 - 4500$, i.e. Rs 2100

Savings for each month in last 3 months = $\frac{2100}{3} = \text{Rs } 700$

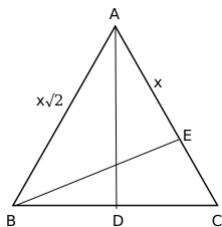
It is given, monthly expenses in last 3 months = Rs 3700

This implies, his monthly earnings from 10th month should be $3700 + 700$, i.e. Rs 4400

The answer is option A.

VIDEO SOLUTION

49.D



It is given, Angle BAE = 45 degrees

This implies $AE = BE$

Let $AE = BE = x$

In right-angled triangle ABD, it is given $\angle ABC = \theta$

$$\sin \theta = \frac{AD}{AB}$$

$$\sin \theta = \frac{AD}{x\sqrt{2}}$$

$$\sqrt{2} \sin \theta = \frac{AD}{BE}$$

The answer is option D.

VIDEO SOLUTION

50.B

$f(x) \geq 0$ for all real numbers x , so $D \leq 0$

Since $f(2)=0$ therefore $x=2$ is a root of $f(x)$

Since the discriminant of $f(x)$ is less than equal to 0 and 2 is a root so we can conclude that $D=0$

Therefore $f(x) = a(x-2)^2$

$f(4)=6$

or, $6 = a(x-2)^2$

$a = 3/2$

$f(-2) = -\frac{3}{2}(-4)^2 = -24$

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51. A

Let the roots of the given equation $5x^3 + cx^2 - 10x + 9 = 0$ be r , $-r$ and p

$$r - r + p = -\frac{c}{5}$$

$$p = -\frac{c}{5} \dots (1)$$

$$-r^2 - pr + pr = -2$$

$$r^2 = 2 \dots (2)$$

$$-r^2 p = -\frac{9}{5}$$

$$p = \frac{9}{10} \dots (3)$$

Substituting p in (1), we get

$$\frac{9}{10} = -\frac{c}{5}$$

$$-\frac{9}{2} = c$$

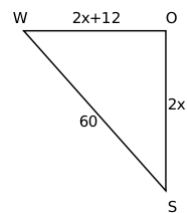
The answer is option A.

[VIDEO SOLUTION](#)

52. C

Let the speeds of two ships be ' x ' and ' $x+6$ ' km per hour

Distance covered in 2 hours will be $2x$ and $2x+12$



It is given,

$$(2x)^2 + (2x+12)^2 = 60^2$$

$$(x)^2 + (x+6)^2 = 30^2$$

$$2x^2 + 12x + 36 = 900$$

$$x^2 + 6x + 18 = 450$$

$$x^2 + 6x - 432 = 0$$

Solving, we get $x = 18$

The speed of slower ship is 18 kmph

The answer is option C.

[VIDEO SOLUTION](#)

53. 12

Given,

$$f(x) + f(x-1) = 1 \dots (1)$$

$$f(x^2 - x) = 5 \dots (2)$$

$$g(x) = x^2$$

Substituting $x = 1$ in (1) and (2), we get

$$f(0) = 5$$

$$f(1) + f(0) = 1$$

$$f(1) = 1 - 5 = -4$$

$$f(2) + f(1) = 1$$

$$f(2) = 1 + 4 = 5$$

$$f(n) = 5 \text{ if } n \text{ is even and } f(n) = -4 \text{ if } n \text{ is odd}$$

$$f(g(5)) + g(f(5)) = f(25) + g(-4) = -4 + 16 = 12$$

[VIDEO SOLUTION](#)

54.24

Let the number of questions attempted be $x+y$ out of which x are correct and y are incorrect and the number of questions unattempted be z .

It is given,

$$x + y + z = 75 \dots\dots (1)$$

$$3x - y + z = 97 \dots\dots (2)$$

$$(2)-(1) \rightarrow x - y = 11$$

$$(1)+(2) \rightarrow 2x + z = 86$$

$$z > x + y$$

$$z > 75 - z$$

$$z > 37.5$$

Minimum possible value of z is 38

$$2x + 38 = 86$$

$$2x = 48$$

$$x = 24$$

The maximum number of correct answers is 24.

[VIDEO SOLUTION](#)

55.10

Let the number of sides of polygons A and B be n and $2n$, respectively.

$$\frac{(n-2)180}{n} = \frac{3}{4}$$

$$\frac{n-2}{n-1} = \frac{3}{4}$$

$$4n - 8 = 3n - 3$$

$$n = 5$$

The number of sides of polygon B is 2×5 , i.e. 10.

[VIDEO SOLUTION](#)

100+ CAT DILR Questions



56.D

Let the number of registered votes be $100x$

The number of votes casted = $80x$

$$\text{Votes received by one of the candidates} = \frac{30}{100} \times 80x = 24x$$

$$\text{Remaining votes} = 80x - 24x = 56x$$

$$\text{Votes received by other three candidates is } \frac{56x}{6}, \frac{2 \times 56x}{6}, \frac{3 \times 56x}{6}$$

It is given,

$$28x - 24x = 2512$$

$$4x = 2512$$

$$x = 628$$

$$\text{The number of registered votes} = 100x = 62800$$

The answer is option D.

[VIDEO SOLUTION](#)

57.A

Given, the number of particles on day 1 = 100

On day 2, one out of every 2 articles produces another particle.

The number of particles on day 2 will be $\frac{100}{2}$, i.e. 50 particles

On day 3, one out of every 3 articles produces another particle.

The number of particles on day 3 will be $\frac{100+50}{3}$, i.e. 50 particles

On day 4, one out of every 4 articles produces another particle.

The number of particles on day 4 will be $\frac{100+50+50}{4}$, i.e. 50 particles

Series will be 100, 50, 50, 50,....

It is given,

$$100 + (m-1)50 = 1000$$

$$m = 19$$

The answer is option A.

[VIDEO SOLUTION](#)

58. **A**

Case 1: 4-digit numbers

Given digits - 0, 1, 2, 3, 4, 5

→ → → →

As the numbers should be greater than 2000, first digit can be 2, 3, 4 and 5.

For remaining digits, we need to arrange 3 digits from the remaining 5 digits, i.e. $5 \times 4 \times 3 = 60$ ways

Total number of possible 4-digit numbers = $4 \times 60 = 240$

Case 2: 5-digit numbers

→ → → → →

First digit cannot be zero.

Therefore, total number of cases = $5 \times 5 \times 4 \times 3 \times 2 = 600$

Case 3: 6-digit numbers

→ → → → → →

First digit cannot be zero.

Therefore, total number of cases = $5 \times 5 \times 4 \times 3 \times 2 \times 1 = 600$

Total number of integers possible = $600 + 600 + 240 = 1440$

The answer is option A.

[VIDEO SOLUTION](#)

59. **B**

To find the largest possible value of n , we need to find the value of n such that $n!$ is less than 15000.

$$7! = 5040$$

$$8! = 40320 > 15000$$

This implies 15000! is not divisible by 40320!

Therefore, maximum value n can take is 7.

The answer is option B.

[VIDEO SOLUTION](#)

60. **6**

Let the time taken by Anu, Tanu and Manu be $5x$, $8x$ and $10x$ hours.

Total work = $\text{LCM}(5x, 8x, 10x) = 40x$

Anu can complete 8 units in one hour

Tanu can complete 5 units in one hour

Manu can complete 4 units in one hour

It is given, three of them together can complete in 32 hours.

$$32(8 + 5 + 4) = 40x$$

$$x = \frac{68}{5}$$

It is given,

Anu and Tanu work together for the first 6 days, working 6 hours 40 minutes per day, i.e. $36 + 4 = 40$ hours

$$40(8 + 5) + y(4) = 40x$$

$$4y = 24$$

$$y = 6$$

Manu alone will complete the remaining work in 6 hours.

[VIDEO SOLUTION](#)



61. D

Initially	1st-Sugar syrup-100L	2nd - Milk - 100L
After step 1	Sugar Syrup - 50L	SS- 50L, Milk - 100L
After step 2	SS - 50+25 = 75L Milk - 50L	SS - 25L, Milk - 50L
After step 3	Milk - 25L SS - 37.5L	Milk - 50 + 25 = 75L SS - 25 + 37.5 = 62.5L

Step 1: Half the content of the first container is transferred to the second container

Step 2: Half of the mixture of second container is transferred back to the first container

Step 3: Half the content of the first container is transferred back to the second container

Sugar syrup : Milk in second container = 62.5 : 75 = 5 : 6

The answer is option D.

VIDEO SOLUTION

62. A

Sum of n terms in an A.P = $\frac{n}{2} (2a + (n-1)d)$

$$A_n = \frac{n}{2} (6 + (n-1)4) = n(2n+1)$$

$$\Sigma A_n = \Sigma n(2n+1) = 2\Sigma n^2 + \Sigma n = \frac{2n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

Substituting n = 25, we get

$$\frac{1}{25} \sum_{n=1}^{25} A_n = \frac{1}{25} \left(\frac{2(25)(25+1)(50+1)}{6} + \frac{25(25+1)}{2} \right)$$

$$\frac{1}{25} \sum_{n=1}^{25} A_n = 26(17) + 13 = 455$$

The answer is option A.

VIDEO SOLUTION

63. 47

Let $\log_2 n = y$

$$\frac{4-y}{3-\frac{y}{2}} < 0$$

$$(4-y) \left(3 - \frac{y}{2} \right) < 0$$

$$(4-y)(6-y) < 0$$

$$(y-4)(y-6) < 0$$

$$4 < y < 6$$

$$4 < \log_2 n < 6$$

$$2^4 < n < 2^6$$

$$16 < n < 64$$

n can take values from 17 to 63(inclusive).

The number of n values possible = 47

VIDEO SOLUTION

64. D

$$a + 2b = 6$$

From the above equation, we can say that maximum value b can take is 3 and minimum value b can take is 0.

$$a + b + b = 6$$

$$a + b = 6 - b$$

a + b is maximum when b is minimum, i.e. b = 0

$$\text{Maximum value of } a + b = 6 - 0 = 6$$

a + b is minimum when b is maximum, i.e. b = 3

$$\text{Minimum value of } a + b = 6 - 3 = 3$$

$$\text{Average} = \frac{6+3}{2} = 4.5$$

The answer is option D.

VIDEO SOLUTION

65.D

The average marks for all the students is 38.

$$\text{Sum} = 5 \times 38 = 190$$

To find the minimum marks scored by Amit, we need to maximise the score of remaining students.

$$\text{Maximum scores sum of remaining students} = 50 + 49 + 48 + 32 = 179$$

$$\text{Minimum possible score of Amit} = 190 - 179 = 11$$

It is given, Amit scored least. This implies maximum possible score of Amit is 31.

$$\text{Difference} = 31 - 11 = 20$$

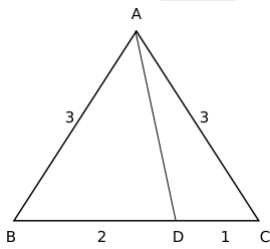
The answer is option D.

[VIDEO SOLUTION](#)

CAT Expected Questions PDF



66.C



Area of triangle ABD is twice the area of triangle ACD

$$\angle ADB = \theta$$

$$\frac{1}{2} (AD) (BD) \sin \theta = 2 \left(\frac{1}{2} (AD) (CD) \sin (180 - \theta) \right)$$

$$BD = 2CD$$

Therefore, $BD = 2$ and $CD = 1$

$$\angle ABC = \angle ACB = 60^\circ$$

Applying cosine rule in triangle ADC, we get

$$\cos \angle ACD = \frac{AC^2 + CD^2 - AD^2}{2(AC)(CD)}$$

$$\frac{1}{2} = \frac{9 + 1 - AD^2}{6}$$

$$AD^2 = 7$$

$$AD = \sqrt{7}$$

The answer is option C.

[VIDEO SOLUTION](#)